END SEMESTER EXAMINATION DECEMBER 2025

Name of the Program: B.Tech. | Semester: V

Name of the Course: Database Management System | Course Code: TCS-503 / API-502

Time: 3 Hours | Maximum Marks: 100

Note:

i. All questions are compulsory.
ii. Answer any two sub questions among a, b & c in each main questions
iii. Total marks in each main question are twenty.
iv. Each questions carries 20 marks

Q1	(20Marks)	
(a)	Discuss Codd's Rules for ~~RDBMS.~~ How do these rules define the foundation and ensure the integrity of a relational database system?	CO1
(b)	Discuss the advantages and disadvantages of using DBMS compared to the file-system approach.	CO1
(c)	Explain the overall architecture of a DBMS. Discuss the role of each component such as query processor, storage manager, transaction manager, and catalog manager.	CO1
Q2		
(a)	What is an Entity Type and Entity Set? Explain the difference between *strong* and *weak entity types* with appropriate diagrams.	CO2
(b)	A university wants to maintain a database for its academic operations. The system should store information about **students, courses, instructors,** and **departments.** Each student has a unique roll number, name, and email. Students can enroll in multiple courses, and each course can have many students. Each course has a unique course code, title, and credit hours, and is offered by a department. Each department has a unique department ID and name. Every instructor belongs to one department and can teach multiple courses. An instructor has an ID, name, and phone number. **Task:** Based on the scenario above: 1. **Design an Entity-Relationship (ER) diagram** showing all the entities, attributes, relationships, and appropriate constraints (like cardinality and participation). 2. Clearly indicate: ○ Weak entities (if any) ○ Primary keys for each entity ○ Multivalued or composite attributes (if any) ○ Relationship types (1:1, 1:N, M:N)	CO2 CO6
(c)	Discuss *super key, candidate key, primary key, alternate key, composite key,* and *foreign key* with suitable examples as well as describe the process of identifying candidate keys for a relation using *functional dependencies* and *attribute closure.*	
Q3		
(a)	Consider the following university database schema and answer the queries using **Relational Algebra:**	CO4
	Schema: • **STUDENT**(*Student_ID*, Name, Dept_ID) • **COURSE**(*Course_ID*, Course_Name, Dept_ID) • **ENROLLMENT**(*Student_ID*, Course_ID) • **INSTRUCTOR**(*Instructor_ID*, Instructor_Name, Dept_ID) • **TEACHES**(*Instructor_ID*, Course_ID) • **DEPARTMENT**(*Dept_ID*, Dept_Name) Write relational algebra expressions for the following queries: (a) Find the names of students enrolled in the course **"DBMS"**. (b) Find the names of instructors who teach **more than one course.** (c) List the names of **departments** that offer **more than three courses.**	CO6 CO3

(b)	Discuss the different types of JOIN operations used in relational algebra. Explain *Theta Join, Equi Join, Natural Join,* and *Outer Join* with examples.	
(c)	(i) Given the relation R(A, B, C, D, E) with the set of functional dependencies $F = \{ A \rightarrow B, B \rightarrow C, CD \rightarrow E, E \rightarrow A \}$, find the **closure of {A, C}+**. (ii) Consider a relation R(P, Q, R, S) with functional dependencies $F = \{ P \rightarrow Q, PQ \rightarrow R, R \rightarrow S \}$. Find the closure of {P}+	
Q4		
(a)	**Write a SQL for Create the following tables with appropriate data types and constraints:** STUDENT(Student_ID, Name, Dept, DOB, Marks) COURSE(Course_ID, Course_Name, Credits) ENROLLMENT(Student_ID, Course_ID, Semester) **Apply the following constraints:** Student_ID → Primary Key Course_ID → Primary Key Student_ID in ENROLLMENT → Foreign Key referencing STUDENT Course_ID in ENROLLMENT → Foreign Key referencing COURSE	CO3 CO5 CO4
(b)	Differentiate between Third Normal Form (3NF) and Boyce–Codd Normal Form (BCNF). Support your answer with a suitable example of a relation that satisfies **3NF** but violates **BCNF.**	
(c)	Given $F=\{A \rightarrow BC, B \rightarrow D, C \rightarrow A, D \rightarrow B\}$: • Find the canonical cover of F. • Identify any redundant dependencies or attributes.	
Q5		
(a)	**Define the ACID properties of a transaction.** Discuss each property with suitable examples.	CO6
(b)	Consider the following concurrent schedule S with two transactions T1 and T2 operating on data items A and B.	CO6
	Time — T_1 — T_2 1 — $R_1(A)$ — 2 — — $R_2(A)$ 3 — $W_1(A)$ — 4 — — $R_2(B)$ 5 — $W_1(B)$ — 6 — — $W_2(B)$ Determine whether the schedule is Conflict serializable or not.	CO5
(c)	Discuss the Two-Phase Locking (2PL) protocol used for concurrency control in DBMS. How does it ensure serializability and recoverability?	